Multivariable Calculus
Problem Set 7: Solutions

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Problem 1 (Positive definite matrix)

(i) Let A be a 2×2 symmetric matrix. Show that A is positive definite if and only if all its diagonal entries and determinant are positive.

Solution. Write

$$A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \qquad a, c \in \mathbb{R},\ b \in \mathbb{R}.$$

A is positive definite iff the quadratic form

$$Q(x, y) = \begin{pmatrix} x & y \end{pmatrix} A \begin{pmatrix} x \\ y \end{pmatrix} = ax^2 + 2bxy + cy^2$$

satisfies $Q(x, y) > 0$ for all $(x, y) \neq (0, 0)$.

($\Rightarrow$) If A is positive definite then

$$a = Q(1, 0) > 0, \qquad c = Q(0, 1) > 0.$$

Also, for each $t \in \mathbb{R}$ we have $0 < Q(1, t) = a + 2bt + ct^2$. This quadratic in t is positive for all t, hence its discriminant is negative:

$$(2b)^2 - 4ac < 0 \quad \Longrightarrow \quad ac - b^2 > 0 \quad \Longrightarrow \quad \det(A) > 0.$$

($\Leftarrow$) Assume $a > 0$, $c > 0$ and $\det(A) = ac - b^2 > 0$. Then

$$Q(x, y) = a\Big(x + \frac{b}{a}y\Big)^2 + \Big(c - \frac{b^2}{a}\Big)y^2.$$

Since $a > 0$ and $c - \frac{b^2}{a} = \frac{ac - b^2}{a} = \frac{\det(A)}{a} > 0$, both terms on the right are nonnegative and they cannot simultaneously vanish unless $(x, y) = (0, 0)$. Hence $Q(x, y) > 0$ for $(x, y) \neq (0, 0)$, so A is positive definite.

(ii) Let A be a 2×2 symmetric matrix. Show that A is negative definite if and only if all its diagonal entries are negative and the determinant is positive.

Solution. A is negative definite iff $-A$ is positive definite. Writing

$$A = \begin{pmatrix} a & b \\ b & c \end{pmatrix},$$

$-A$ is positive definite iff $-a > 0$ and $\det(-A) > 0$. But $\det(-A) = \det(A)$ for 2×2 matrices, so we need $a < 0$ and $\det(A) > 0$. Moreover, if $a < 0$ and $\det(A) = ac - b^2 > 0$, then $ac > b^2 \geq 0$ forces $ac > 0$, and since $a < 0$ we get $c < 0$. Thus the conditions are equivalent to "both diagonal entries negative and $\det(A) > 0$".

(iii) Let A be an $n \times n$ positive definite matrix. Show that all its diagonal entries are positive.

Solution. Let $e_i \in \mathbb{R}^n$ be the ith standard basis vector. Positive definiteness means $v^\top A v > 0$ for all $v \neq 0$. Taking $v = e_i \neq 0$ gives

$$e_i^\top A e_i = a_{ii} > 0.$$

Hence every diagonal entry a_{ii} is positive.

Problem 2

Find the local maximum and minimum values and saddle point(s) of the following functions.

(i) $f(x, y) = x^2 + xy + y^2 + y$.

Solution.

$$f_x = 2x + y, \qquad f_y = x + 2y + 1.$$

Solving $f_x = f_y = 0$ gives $y = -2x$ and $x - 4x + 1 = 0$, hence

$$(x, y) = \left(\frac{1}{3}, -\frac{2}{3}\right).$$

The Hessian is

$$H = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix},$$

which is positive definite ($\det(H) = 3 > 0$ and $2 > 0$). Therefore the critical point is a strict local minimum (in fact global minimum since f is a strictly convex quadratic). The minimum value is

$$f\left(\frac{1}{3}, -\frac{2}{3}\right) = \frac{1}{9} - \frac{2}{9} + \frac{4}{9} - \frac{2}{3} = -\frac{1}{3}.$$

(ii) $f(x, y) = x^3 + y^2 - 3x$.

Solution.

$$f_x = 3x^2 - 3, \qquad f_y = 2y.$$

Hence critical points satisfy $y = 0$ and $x = \pm 1$: the points are $(1, 0)$ and $(-1, 0)$. The Hessian is diagonal:

$$f_{xx} = 6x, \qquad f_{yy} = 2, \qquad f_{xy} = 0.$$

At $(1, 0)$ we have $f_{xx} = 6 > 0$ and $f_{yy} = 2 > 0$, so H is positive definite and $(1, 0)$ is a strict local minimum. Its value is $f(1, 0) = 1 - 3 = -2$.

At $(-1, 0)$ we have $f_{xx} = -6 < 0$ and $f_{yy} = 2 > 0$, so the Hessian is indefinite and $(-1, 0)$ is a saddle point. Its value is $f(-1, 0) = -1 + 3 = 2$.

(iii) $f(x,y) = (x-y)(1-xy)$.

Solution. Expand $f = x - y - x^2y + xy^2$. Then

$$f_x = 1 - 2xy + y^2, \qquad f_y = -1 - x^2 + 2xy.$$

Setting $f_x = f_y = 0$ gives

$$y^2 - 2xy + 1 = 0, \qquad 2xy - x^2 - 1 = 0.$$

Using $2xy = x^2 + 1$ in the first equation yields $y^2 - x^2 = 0$, so $y = \pm x$. If $y = x$, then $2x^2 - x^2 - 1 = 0$ gives $x^2 = 1$, so $(x,y) = (1,1), (-1,-1)$. If $y = -x$, then $2x(-x) - x^2 - 1 = -3x^2 - 1 = 0$ has no real solution. Hence the only critical points are $(1,1)$ and $(-1,-1)$.

The second derivatives are

$$f_{xx} = -2y, \qquad f_{yy} = 2x, \qquad f_{xy} = -2x + 2y.$$

At $(1,1)$, $f_{xx} = -2$, $f_{yy} = 2$, $f_{xy} = 0$, so $\det(H) = -4 < 0$ and $(1,1)$ is a saddle point. Similarly at $(-1,-1)$, $f_{xx} = 2$, $f_{yy} = -2$, $f_{xy} = 0$, so $\det(H) = -4 < 0$ and $(-1,-1)$ is a saddle point. (Both have value $f = 0$ since $x - y = 0$.)

(iv) $f(x,y) = y^3 + 3x^2y - 6x^2 - 6y^2 + 2$.

Solution.

$$f_x = 6xy - 12x = 6x(y-2), \qquad f_y = 3y^2 + 3x^2 - 12y = 3(x^2 + y^2 - 4y).$$

So $f_x = 0$ gives $x = 0$ or $y = 2$. Also $f_y = 0$ gives $x^2 + y^2 - 4y = 0$, i.e.

$$x^2 + (y-2)^2 = 4.$$

Intersections yield the critical points

$$(0,0),\ (0,4),\ (2,2),\ (-2,2).$$

The Hessian is

$$f_{xx} = 6y - 12, \quad f_{yy} = 6y - 12, \quad f_{xy} = 6x.$$

At $(0,0)$: $H = \mathrm{diag}(-12,-12)$ is negative definite, so $(0,0)$ is a strict local maximum with value $f(0,0) = 2$.

At $(0,4)$: $H = \mathrm{diag}(12,12)$ is positive definite, so $(0,4)$ is a strict local minimum with value $f(0,4) = 64 - 96 + 2 = -30$.

At $(2,2)$: $H = \left(\begin{smallmatrix} 0 & 12 \\ 12 & 0 \end{smallmatrix}\right)$ is indefinite, so $(2,2)$ is a saddle; $f(2,2) = 8 + 24 - 24 - 24 + 2 = -14$.
At $(-2,2)$: $H = \left(\begin{smallmatrix} 0 & -12 \\ -12 & 0 \end{smallmatrix}\right)$ is indefinite, so $(-2,2)$ is also a saddle, with the same value -14.

(v) $f(x,y) = xy(1-x-y)$.

Solution. Expand $f = xy - x^2y - xy^2$. Then

$$f_x = y - 2xy - y^2 = y(1-2x-y), \qquad f_y = x - x^2 - 2xy = x(1-x-2y).$$

Critical points satisfy $f_x = f_y = 0$. From the factorizations:

$$y = 0 \text{ or } y = 1 - 2x, \qquad x = 0 \text{ or } x = 1 - 2y.$$

This gives the points

$$(0,0),\ (1,0),\ (0,1),\ \left(\frac{1}{3},\frac{1}{3}\right).$$

Second derivatives:

$$f_{xx} = -2y, \qquad f_{yy} = -2x, \qquad f_{xy} = 1 - 2x - 2y.$$

At $(\frac{1}{3},\frac{1}{3})$ we have $f_{xx} = f_{yy} = -\frac{2}{3}$ and $f_{xy} = -\frac{1}{3}$, hence

$$\det(H) = \frac{4}{9} - \frac{1}{9} = \frac{1}{3} > 0, \quad f_{xx} < 0,$$

so $(\frac{1}{3},\frac{1}{3})$ is a strict local maximum with value $f = \frac{1}{27}$. At $(0,0)$, $(1,0)$, $(0,1)$ one finds $\det(H) < 0$, so these are saddle points (in $\mathbb{R}^2$).

(vi) $f(x,y) = e^x \cos y$.

Solution.

$$f_x = e^x \cos y, \qquad f_y = -e^x \sin y.$$

Since $e^x \neq 0$, any critical point would need $\cos y = 0$ and $\sin y = 0$ simultaneously, which is impossible. Hence there are *no* critical points, and thus no local maxima/minima/saddles in $\mathbb{R}^2$.

(vii) $f(x,y) = (x^2+y^2)e^{y^2-x^2}$.

Solution. Let $r^2 = x^2+y^2$ and $E = e^{y^2-x^2}$. Then

$$f = r^2 E, \qquad f_x = 2xE + r^2(-2xE) = 2xE(1-r^2),$$

$$f_y = 2yE + r^2(2yE) = 2yE(1+r^2).$$

Since $E > 0$ and $1 + r^2 > 0$, the equation $f_y = 0$ forces $y = 0$. Then $f_x = 0$ becomes $2xe^{-x^2}(1-x^2) = 0$, so $x = 0$ or $x = \pm 1$. Thus the critical points are $(0,0)$, $(1,0)$, $(-1,0)$.

For classification, compute

$$f_{xx} = E\big(2 - 10x^2 - 2y^2 + 4x^4 + 4x^2y^2\big), \quad f_{yy} = E\big(2 + 2x^2 + 10y^2 + 4x^2y^2 + 4y^4\big),$$

$$f_{xy} = -4xyr^2E.$$

At $(0,0)$: $H = \mathrm{diag}(2,2)$ is positive definite, so $(0,0)$ is a strict local minimum with value $f(0,0) = 0$.

At $(\pm 1,0)$: $f_{xy} = 0$, $f_{xx} = -4e^{-1} < 0$, $f_{yy} = 4e^{-1} > 0$, so $\det(H) < 0$ and $(1,0),(-1,0)$ are saddle points.

(viii) $f(x,y) = e^y(y^2 - x^2)$.

Solution.

$$f_x = -2xe^y, \qquad f_y = e^y(y^2 - x^2 + 2y).$$

Critical points satisfy $x = 0$ and $y^2 + 2y = 0$, so $y = 0$ or $y = -2$. Hence the critical points are $(0,0)$ and $(0,-2)$.

Second derivatives:

$$f_{xx} = -2e^y, \qquad f_{xy} = -2xe^y, \qquad f_{yy} = e^y(y^2 - x^2 + 4y + 2).$$

At $(0,0)$: $f_{xx} = -2$, $f_{yy} = 2$, $f_{xy} = 0$, so $\det(H) = -4 < 0$ and $(0,0)$ is a saddle point. At $(0,-2)$: $f_{xx} = -2e^{-2} < 0$, $f_{yy} = -2e^{-2} < 0$, $\det(H) = 4e^{-4} > 0$, so $(0,-2)$ is a strict local maximum with value $f(0,-2) = 4e^{-2}$.

(ix) $f(x,y) = y^2 - 2y\cos x, \quad 1 \le x \le 7.$

Solution. First note the useful rewriting

$$f(x,y) = (y - \cos x)^2 - \cos^2 x,$$

so for each fixed x the function is a parabola in y with minimum at $y = \cos x$.

In the interior $1 < x < 7$, critical points satisfy

$$f_x = 2y\sin x = 0, \qquad f_y = 2y - 2\cos x = 0.$$

Hence $y = \cos x$, and then $f_x = 2\cos x \sin x = \sin(2x) = 0$. Therefore interior critical points occur at

$$x \in \left\{\frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi\right\} \cap (1,7), \quad y = \cos x,$$

i.e.

$$\left(\frac{\pi}{2}, 0\right), \ (\pi, -1), \ \left(\frac{3\pi}{2}, 0\right), \ (2\pi, 1).$$

The Hessian is

$$f_{xx} = 2y\cos x, \qquad f_{yy} = 2, \qquad f_{xy} = 2\sin x.$$

At $x = \pi$ and $x = 2\pi$ we have $\cos x = \pm 1$, $\sin x = 0$, and $y = \cos x$, so

$$f_{xx} = 2\cos^2 x = 2 > 0, \quad f_{xy} = 0, \quad f_{yy} = 2,$$

hence these are strict local minima (relative to the strip) at $(\pi, -1)$ and $(2\pi, 1)$. At $x = \frac{\pi}{2}$ or $x = \frac{3\pi}{2}$ we have $\cos x = 0$, $y = 0$, and $\sin x = \pm 1$, so $\det(H) < 0$ and these are saddle points.

On the boundary $x = 1$ or $x = 7$, f as a function of y still has a (one-dimensional) minimum at $y = \cos x$, giving boundary minima in the strip at $(1, \cos 1)$ and $(7, \cos 7)$.

(x) $f(x,y) = \sin x \, \sin y, \quad -\pi \le x \le \pi, \ -\pi \le y \le \pi.$

Solution.

$$f_x = \cos x \, \sin y, \qquad f_y = \sin x \, \cos y.$$

Thus critical points satisfy $\cos x \, \sin y = 0$ and $\sin x \, \cos y = 0$. There are two families of solutions:

- $\sin x = 0$ and $\sin y = 0$, i.e. $x \in \{-\pi, 0, \pi\}$ and $y \in \{-\pi, 0, \pi\}$ (9 points), where $f = 0$.
- $\cos x = 0$ and $\cos y = 0$, i.e. $x \in \{-\frac{\pi}{2}, \frac{\pi}{2}\}$ and $y \in \{-\frac{\pi}{2}, \frac{\pi}{2}\}$ (4 points), where $f = \pm 1$.

The Hessian entries are

$$f_{xx} = -\sin x \sin y = -f, \qquad f_{yy} = -\sin x \sin y = -f, \qquad f_{xy} = \cos x \cos y.$$

At points with $x = \pm\frac{\pi}{2}$ and $y = \pm\frac{\pi}{2}$, we have $f_{xy} = 0$ and $f = \pm 1$. Hence when $f = 1$ (same signs), $H = -I$ is negative definite, giving strict local maxima with value 1 at

$$\left(\frac{\pi}{2}, \frac{\pi}{2}\right), \quad \left(-\frac{\pi}{2}, -\frac{\pi}{2}\right).$$

When $f = -1$ (opposite signs), $H = I$ is positive definite, giving strict local minima with value -1 at

$$\left(\frac{\pi}{2}, -\frac{\pi}{2}\right), \quad \left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$$

At the 9 points where $\sin x = \sin y = 0$, we have $f = 0$ and $f_{xx} = f_{yy} = 0$ while $f_{xy} = \cos x \cos y = \pm 1$, so $\det(H) < 0$ and these are saddle points (relative to $\mathbb{R}^2$, and also relative to the square except for edge effects at the boundary).

Problem 3

(i) Find all local and global extreme values for $f(x, y) = xy(1 - x^2 - y^2)$ on $(x, y) \in [0, 1]^2$.

Solution. On $\mathbb{R}^2$,

$$f(x, y) = xy - x^3y - xy^3.$$

Compute

$$f_x = y - 3x^2y - y^3 = y(1 - 3x^2 - y^2), \qquad f_y = x - x^3 - 3xy^2 = x(1 - x^2 - 3y^2).$$

Interior critical points. In $(0, 1)^2$ we have $x > 0, y > 0$, so we solve

$$1 - 3x^2 - y^2 = 0, \qquad 1 - x^2 - 3y^2 = 0.$$

Subtracting gives $-2x^2 + 2y^2 = 0$, hence $x^2 = y^2$ and with $x, y > 0$ we get $x = y$. Plugging into $1 - 3x^2 - x^2 = 0$ yields $x^2 = \frac{1}{4}$, so $(x, y) = (\frac{1}{2}, \frac{1}{2})$.

The Hessian components are

$$f_{xx} = -6xy, \qquad f_{yy} = -6xy, \qquad f_{xy} = 1 - 3x^2 - 3y^2.$$

At $(\frac{1}{2}, \frac{1}{2})$:

$$f_{xx} = f_{yy} = -\frac{3}{2}, \quad f_{xy} = -\frac{1}{2}, \quad \det(H) = \frac{9}{4} - \frac{1}{4} = 2 > 0,$$

and $f_{xx} < 0$, so $(\frac{1}{2}, \frac{1}{2})$ is a strict local maximum. Its value is

$$f\left(\frac{1}{2}, \frac{1}{2}\right) = \frac{1}{4}\left(1 - \frac{1}{4} - \frac{1}{4}\right) = \frac{1}{8}.$$

Boundary analysis. On $x = 0$ or $y = 0$, we have $f = 0$. On $x = 1$, $f(1, y) = y(1 - 1 - y^2) = -y^3$, whose minimum on $[0, 1]$ is -1 at $y = 1$ and maximum is 0 at $y = 0$. On $y = 1$, $f(x, 1) = -x^3$, whose minimum on $[0, 1]$ is -1 at $x = 1$ and maximum is 0 at $x = 0$.

Global extrema on $[0, 1]^2$. Comparing interior and boundary candidates:

- Global maximum: $\max_{[0,1]^2} f = \dfrac{1}{8}$, attained at $(\frac{1}{2}, \frac{1}{2})$.
- Global minimum: $\min_{[0,1]^2} f = -1$, attained at $(1, 1)$.

Also, $f = 0$ occurs on the axes and along the quarter-circle $x^2 + y^2 = 1$ inside the square.

(ii) Let $f(x, y) = 3x^4 - 4x^2y + y^2$. Show that on every line $y = mx$, the function has a minimum at $(0, 0)$, but $(0, 0)$ is not a local minimum.

Solution. Restrict to the line $y = mx$:

$$f(x, mx) = 3x^4 - 4x^2(mx) + (mx)^2 = 3x^4 - 4mx^3 + m^2x^2 = x^2(3x^2 - 4mx + m^2).$$

Since $x^2 \geq 0$ and $3x^2 - 4mx + m^2$ is continuous with value $m^2 \geq 0$ at $x = 0$, we have $f(x, mx) \geq 0$ for all x sufficiently close to 0, and $f(0, 0) = 0$. In particular, along each fixed line $y = mx$, the restricted one-variable function $x \mapsto f(x, mx)$ has a *local* minimum at $x = 0$ (with local minimum value 0). (This minimum need not be global along the entire line.)

However, $(0, 0)$ is *not* a local minimum in $\mathbb{R}^2$. Indeed, along the curve $y = 2x^2$ we get

$$f(x, 2x^2) = 3x^4 - 4x^2(2x^2) + (2x^2)^2 = 3x^4 - 8x^4 + 4x^4 = -x^4 < 0$$

for all $x \neq 0$. So in every neighborhood of $(0, 0)$ there are points with $f(x, y) < f(0, 0) = 0$, proving $(0, 0)$ is not a local minimum.

(iii) Given n distinct numbers $a_1, \ldots, a_n \in \mathbb{R}$ and real numbers $b_1, \ldots, b_n \in \mathbb{R}$, find the straight line $y = ax + b$ which minimizes

$$\sum_{i=1}^{n} \big(ax_i + b - b_i\big)^2$$

in the special case where the "x-coordinates" are $x_i = a_i$ (as encoded in the problem set), i.e. minimize

$$F(b, a) = \sum_{i=1}^{n} \big(b + a\, a_i - b_i\big)^2.$$

Solution. Define

$$F(b, a) = \sum_{i=1}^{n} (b + aa_i - b_i)^2, \qquad (b, a) \in \mathbb{R}^2.$$

This is a strictly convex quadratic in (b, a) as long as not all a_i are equal; since the a_i are distinct, it has a unique global minimizer given by the normal equations $\nabla F = 0$.

Compute partial derivatives:

$$\frac{\partial F}{\partial b} = 2\sum_{i=1}^{n} (b + aa_i - b_i), \qquad \frac{\partial F}{\partial a} = 2\sum_{i=1}^{n} a_i(b + aa_i - b_i).$$

Setting them to zero gives the linear system

$$nb + a\sum_{i=1}^{n} a_i = \sum_{i=1}^{n} b_i, \qquad b\sum_{i=1}^{n} a_i + a\sum_{i=1}^{n} a_i^2 = \sum_{i=1}^{n} a_i b_i.$$

Let

$$S_a = \sum_{i=1}^{n} a_i, \quad S_{aa} = \sum_{i=1}^{n} a_i^2, \quad S_b = \sum_{i=1}^{n} b_i, \quad S_{ab} = \sum_{i=1}^{n} a_i b_i.$$

The determinant is

$$D = nS_{aa} - S_a^2.$$

Since the a_i are not all equal, $D > 0$ (it equals n^2 times the variance of the a_i). Solving,

$$a^\star = \frac{nS_{ab} - S_a S_b}{D}, \qquad b^\star = \frac{S_b S_{aa} - S_a S_{ab}}{D}.$$

Therefore the minimizing line is $y = a^\star x + b^\star$.

Problem 4

Consider $f(x, y) = xy(1 - x^2 - y^2)$ on $[0, 1]^2$.

(i) Find all the critical points on the square $[0, 1]^2$.

Solution. As in Problem 3(i),

$$f_x = y(1 - 3x^2 - y^2), \qquad f_y = x(1 - x^2 - 3y^2).$$

By the standard definition, critical points are interior points where $\nabla f = 0$. Thus we work on $(0, 1)^2$. Since $x > 0$ and $y > 0$ in $(0, 1)^2$, the equations $f_x = 0$ and $f_y = 0$ reduce to

$$1 - 3x^2 - y^2 = 0, \qquad 1 - x^2 - 3y^2 = 0.$$

Subtracting the second equation from the first gives

$$-2x^2 + 2y^2 = 0 \quad \Longrightarrow \quad x^2 = y^2.$$

Because $x, y > 0$ in the interior of the square, this implies $x = y$. Substituting $y = x$ into $1 - 3x^2 - y^2 = 0$, we get

$$1 - 3x^2 - x^2 = 0 \quad \Longrightarrow \quad 1 - 4x^2 = 0 \quad \Longrightarrow \quad x^2 = \frac{1}{4}.$$

Again using $x > 0$, we obtain $x = \frac{1}{2}$, and hence $y = \frac{1}{2}$. Therefore the unique critical point in $(0, 1)^2$ is

$$\left(\frac{1}{2}, \frac{1}{2}\right).$$

Boundary points are handled separately when finding extrema on $[0, 1]^2$ (see parts (iii)–(iv)).

(ii) Find all saddle points, points of local minima and points of local maxima on $(0, 1)^2$.

Solution. From part (i), the only critical point in $(0, 1)^2$ is $(\frac{1}{2}, \frac{1}{2})$. To classify it, we compute the second derivatives:

$$f_{xx} = -6xy, \qquad f_{yy} = -6xy, \qquad f_{xy} = 1 - 3x^2 - 3y^2.$$

At $\left(\frac{1}{2}, \frac{1}{2}\right)$ these become

$$f_{xx}\left(\frac{1}{2}, \frac{1}{2}\right) = -6 \cdot \frac{1}{2} \cdot \frac{1}{2} = -\frac{3}{2},$$
$$f_{yy}\left(\frac{1}{2}, \frac{1}{2}\right) = -\frac{3}{2},$$
$$f_{xy}\left(\frac{1}{2}, \frac{1}{2}\right) = 1 - 3 \cdot \frac{1}{4} - 3 \cdot \frac{1}{4} = 1 - \frac{3}{2} = -\frac{1}{2}.$$

Hence the Hessian determinant is

$$D = f_{xx} f_{yy} - (f_{xy})^2 = \left(-\frac{3}{2}\right)\left(-\frac{3}{2}\right) - \left(-\frac{1}{2}\right)^2 = \frac{9}{4} - \frac{1}{4} = 2 > 0.$$

Since $D > 0$ and $f_{xx}\left(\frac{1}{2}, \frac{1}{2}\right) = -\frac{3}{2} < 0$, the second derivative test shows that $\left(\frac{1}{2}, \frac{1}{2}\right)$ is a strict local maximum. There are no interior local minima or saddle points.

(iii) Find all saddle points, points of local minima and points of local maxima on $[0, 1]^2$.

Solution (relative to the square). We now classify points *relative to the square* $[0, 1]^2$.

First, from part (ii), the only interior critical point is $\left(\frac{1}{2}, \frac{1}{2}\right)$, and it is a strict local maximum.

Next consider boundary points. Since

$$f(x, y) = xy(1 - x^2 - y^2),$$

we have

$$f(x, 0) = 0, \qquad f(0, y) = 0$$

for all $x, y \in [0, 1]$. If $(x_0, 0)$ is a point on the bottom edge with $0 < x_0 < 1$, then for nearby points $(x, y) \in [0, 1]^2$ we have $x \geq 0$, $y \geq 0$, and if $x_0 < 1$ then for sufficiently small perturbations we also have $x^2 + y^2 < 1$. Hence

$$1 - x^2 - y^2 > 0$$

near $(x_0, 0)$, so

$$f(x, y) = xy(1 - x^2 - y^2) \geq 0 = f(x_0, 0).$$

Therefore every point $(x_0, 0)$ with $0 \leq x_0 < 1$ is a local minimum relative to the square. The same argument shows that every point $(0, y_0)$ with $0 \leq y_0 < 1$ is also a local minimum relative to the square.

In particular, $(0, 0)$ is a local minimum.

At the corner $(1, 1)$ we have

$$f(1, 1) = 1 \cdot 1 \cdot (1 - 1 - 1) = -1.$$

For points (x, y) near $(1, 1)$ inside the square, we have $x > 0$, $y > 0$, and

$$1 - x^2 - y^2 < 0,$$

so $f(x, y) < 0$. Since the global minimum will turn out to be -1, this shows $(1, 1)$ is a boundary local minimum, in fact the global minimum.

There are no boundary local maxima: on the axes the value is 0, but moving slightly into the interior near points with $x^2 + y^2 < 1$ makes $f > 0$, so such points cannot be local maxima. Also near $(1, 1)$ the nearby values are larger than -1, so $(1, 1)$ is a minimum, not a maximum.

Thus, relative to $[0, 1]^2$:

- the only local maximum is $\left(\frac{1}{2}, \frac{1}{2}\right)$;
- the local minima are all points on the two edges $y = 0$ and $x = 0$ with $x^2 + y^2 < 1$, together with the corner $(1, 1)$;
- there are no saddle points in the interior, and the only interior stationary point is the local maximum already found.

(iv) Find maxima and minima (point of attainment as well as value) on $[0, 1]^2$.

Solution. Since $[0, 1]^2$ is compact and f is continuous, absolute maxima and minima exist. From Problem 3(i), the only interior critical point is $\left(\frac{1}{2}, \frac{1}{2}\right)$, where

$$f\left(\frac{1}{2}, \frac{1}{2}\right) = \frac{1}{2} \cdot \frac{1}{2} \cdot \left(1 - \frac{1}{4} - \frac{1}{4}\right) = \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}.$$

Now examine the boundary:

On $x = 0$ or $y = 0$,

$$f = 0.$$

On $x = 1$,

$$f(1, y) = y(1 - 1 - y^2) = -y^3, \qquad 0 \le y \le 1.$$

This is decreasing on $[0, 1]$, so along this edge its maximum is 0 at $y = 0$ and its minimum is -1 at $y = 1$. Similarly, on $y = 1$,

$$f(x, 1) = x(1 - x^2 - 1) = -x^3, \qquad 0 \le x \le 1,$$

whose maximum is 0 at $x = 0$ and minimum is -1 at $x = 1$.

Comparing the interior value $\frac{1}{8}$ with all boundary values,

$$\max_{[0,1]^2} f = \frac{1}{8} \text{ at } \left(\frac{1}{2}, \frac{1}{2}\right), \qquad \min_{[0,1]^2} f = -1 \text{ at } (1, 1).$$

Problem 5

Find extreme values for the following function subject to the given constraint.

(i) $f(x, y) = xy$ subject to $x + y = 1$.

Solution. Use the constraint

$$g(x, y) = x + y - 1 = 0.$$

Then

$$\nabla f = (y, x), \qquad \nabla g = (1, 1).$$

The Lagrange multiplier equation $\nabla f = \lambda \nabla g$ gives

$$(y, x) = \lambda(1, 1),$$

so

$$y = \lambda, \qquad x = \lambda.$$

Hence $x = y$. Using the constraint $x + y = 1$, we get

$$2x = 1 \quad \Longrightarrow \quad x = \frac{1}{2},$$

and therefore

$$y = \frac{1}{2}.$$

So the only critical point on the constraint is $\left(\frac{1}{2}, \frac{1}{2}\right)$, and

$$f\left(\frac{1}{2}, \frac{1}{2}\right) = \frac{1}{4}.$$

To show that this is the maximum, we may now use the one-variable form of the constraint:

$$y = 1 - x \quad \Longrightarrow \quad f(x, 1-x) = x(1-x) = -\left(x - \frac{1}{2}\right)^2 + \frac{1}{4} \leq \frac{1}{4}.$$

Thus the maximum value is $\frac{1}{4}$, attained at $\left(\frac{1}{2}, \frac{1}{2}\right)$.

To see that there is no minimum, let $x \to +\infty$ along the line $x+y = 1$. Then $y = 1-x \to -\infty$ and

$$f(x, 1-x) = x(1-x) = x - x^2 \to -\infty.$$

There is no minimum (the function is unbounded below along the line $x + y = 1$).

(ii) $f(x, y) = x^2 + y^2$ subject to $\frac{x}{a} + \frac{y}{b} = 1$.

Solution. Assume $a, b \neq 0$. Use Lagrange multipliers with $g(x, y) = \frac{x}{a} + \frac{y}{b} - 1 = 0$. Then

$$\nabla f = (2x, 2y), \qquad \nabla g = \left(\frac{1}{a}, \frac{1}{b}\right),$$

so $(2x, 2y) = \lambda(\frac{1}{a}, \frac{1}{b})$, giving

$$x = \frac{\lambda}{2a}, \qquad y = \frac{\lambda}{2b}.$$

Plug into the constraint:

$$\frac{1}{a} \cdot \frac{\lambda}{2a} + \frac{1}{b} \cdot \frac{\lambda}{2b} = 1 \quad \Longrightarrow \quad \lambda\left(\frac{1}{2a^2} + \frac{1}{2b^2}\right) = 1 \quad \Longrightarrow \quad \lambda = \frac{2a^2b^2}{a^2 + b^2}.$$

Hence

$$x^\star = \frac{ab^2}{a^2 + b^2}, \qquad y^\star = \frac{a^2b}{a^2 + b^2}.$$

The minimum value is

$$(x^\star)^2 + (y^\star)^2 = \frac{a^2b^4 + a^4b^2}{(a^2 + b^2)^2} = \frac{a^2b^2}{a^2 + b^2}.$$

To see there is no maximum, parametrize the constraint line by

$$y = b - \frac{b}{a}x.$$

Then

$$x^2 + y^2 = x^2 + \left(b - \frac{b}{a}x\right)^2.$$

As $|x| \to \infty$, the right-hand side tends to $+\infty$, so the function is unbounded above on the constraint line. Hence no maximum exists.

(iii) $f(x,y) = x + y$ subject to $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Solution. Use Lagrange multipliers with $g(x,y) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - 1 = 0$. Then

$$(1,1) = \lambda\Big(\frac{2x}{a^2}, \frac{2y}{b^2}\Big),$$

so

$$x = \frac{a^2}{2\lambda}, \qquad y = \frac{b^2}{2\lambda}.$$

Plugging into the constraint:

$$\frac{1}{a^2} \cdot \frac{a^4}{4\lambda^2} + \frac{1}{b^2} \cdot \frac{b^4}{4\lambda^2} = 1 \quad \Longrightarrow \quad \frac{a^2+b^2}{4\lambda^2} = 1 \quad \Longrightarrow \quad \lambda = \pm\frac{1}{2}\sqrt{a^2+b^2}.$$

Therefore

$$(x,y) = \pm\Big(\frac{a^2}{\sqrt{a^2+b^2}}, \frac{b^2}{\sqrt{a^2+b^2}}\Big).$$

The corresponding values are

$$f_{\max} = \sqrt{a^2+b^2}, \qquad f_{\min} = -\sqrt{a^2+b^2}.$$

Concretely,

$$f\left(\frac{a^2}{\sqrt{a^2+b^2}}, \frac{b^2}{\sqrt{a^2+b^2}}\right) = \frac{a^2+b^2}{\sqrt{a^2+b^2}} = \sqrt{a^2+b^2},$$

while choosing the negative point gives the negative of this value.

Problem 6 (Lagrange multipliers)

Use Lagrange multipliers to find the maximum and minimum of the given function subject to the given constraint.

(i) $f(x,y) = x^2 + y^2, \quad g(x,y) = xy - 1 = 0.$

Solution.

$$(2x, 2y) = \lambda(y, x).$$

Thus

$$2x = \lambda y, \qquad 2y = \lambda x.$$

Multiplying these two equations gives

$$4xy = \lambda^2 xy.$$

Since $xy = 1 \neq 0$, $\lambda^2 = 4$, so $\lambda = \pm 2$. If $\lambda = 2$, then $x = y$; with $xy = 1$ this yields $(1,1)$ and $(-1,-1)$. If $\lambda = -2$, then $x = -y$, which contradicts $xy = 1$. Therefore the only candidates are $(1,1)$ and $(-1,-1)$, and both give $f = 2$. Moreover, on the constraint $y = 1/x$,

$$f(x, 1/x) = x^2 + \frac{1}{x^2} \geq 2$$

by AM–GM, so $f_{\min} = 2$ at $(1,1)$ and $(-1,-1)$. There is no maximum (unbounded above).

(ii) $f(x,y) = x^2 + 2y^2 - 4y, \quad g(x,y) = x^2 + y^2 - 9 = 0.$

Solution.

$$(2x, 4y-4) = \lambda(2x, 2y).$$

So

$$2x = 2\lambda x, \qquad 4y - 4 = 2\lambda y.$$

From the first equation,

$$2x(1-\lambda) = 0,$$

hence either $x = 0$ or $\lambda = 1$. If $\lambda = 1$, then $4y - 4 = 2y$, so $y = 2$, and $x^2 = 9 - 4 = 5$, so $(x,y) = (\pm\sqrt{5}, 2)$ with $f = 5$. If $x = 0$, then $y = \pm 3$. Values:

$$f(0,3) = 18 - 12 = 6, \qquad f(0,-3) = 18 + 12 = 30.$$

Hence on the circle $x^2+y^2 = 9$, the global minimum is 5 at $(\pm\sqrt{5}, 2)$ and the global maximum is 30 at $(0,-3)$.

(iii) $f(x,y) = y^2 - x^2, \quad g(x,y) = \frac{1}{x^2} + y^2 - 1 = 0.$

Solution.

$$(-2x, 2y) = \lambda\Big(-\frac{2}{x^3}, 2y\Big).$$

Thus

$$-2x = -\frac{2\lambda}{x^3}, \qquad 2y = 2\lambda y.$$

From the second component,

$$2y(1-\lambda) = 0,$$

so either $y = 0$ or $\lambda = 1$. If $\lambda = 1$, then $-2x = -2/x^3$, so $x^4 = 1$ and $x = \pm 1$. The constraint then gives $y = 0$. If $y = 0$, the constraint is $1/x^2 = 1$, again giving $x = \pm 1$. Thus the only candidates are $(1,0)$ and $(-1,0)$, at which $f = -1$. On the constraint, $y^2 = 1 - \frac{1}{x^2}$ and

$$f = 1 - \frac{1}{x^2} - x^2 = 1 - \Big(x^2 + \frac{1}{x^2}\Big) \le -1$$

with equality exactly at $|x| = 1$. Hence the global maximum is -1 at $(\pm 1, 0)$; there is no minimum (unbounded below as $|x| \to \infty$).

(iv) $f(x,y) = xy, \quad g(x,y) = x^2 + 2y^2 - 6 = 0.$

Solution.

$$(y, x) = \lambda(2x, 4y).$$

So

$$y = 2\lambda x, \qquad x = 4\lambda y.$$

Substitute the first equation into the second:

$$x = 4\lambda(2\lambda x) = 8\lambda^2 x.$$

If $x = 0$, then from the constraint $2y^2 = 6$, so $y = \pm\sqrt{3}$, and then $f = xy = 0$. These are not extrema once compared with the nonzero values found below. If $x \neq 0$, we may divide by x and get

$$8\lambda^2 = 1 \quad \Longrightarrow \quad \lambda = \pm\frac{1}{2\sqrt{2}}.$$

Then $y = \pm\frac{x}{\sqrt{2}}$ (sign matches λ). Plug into the constraint:

$$x^2 + 2 \cdot \frac{x^2}{2} = 2x^2 = 6 \quad \Longrightarrow \quad x = \pm\sqrt{3},$$

hence $y = \pm\sqrt{\frac{3}{2}}$ with either same sign (for $\lambda > 0$) or opposite sign (for $\lambda < 0$). Values:

$$f_{\max} = \frac{3}{\sqrt{2}} \text{ at } (\sqrt{3}, \sqrt{3/2}),\ (-\sqrt{3}, -\sqrt{3/2}),$$

$$f_{\min} = -\frac{3}{\sqrt{2}} \text{ at } (\sqrt{3}, -\sqrt{3/2}),\ (-\sqrt{3}, \sqrt{3/2}).$$

(v) $f(x, y) = x^2y^2, \quad g(x, y) = x^2 + y^2 - 1 = 0.$

Solution.

$$(2xy^2, 2yx^2) = \lambda(2x, 2y).$$

Thus

$$2xy^2 = 2\lambda x, \qquad 2yx^2 = 2\lambda y.$$

If $x = 0$ or $y = 0$, the constraint gives $(\pm 1, 0)$ or $(0, \pm 1)$ and then

$$f = x^2y^2 = 0.$$

If $x \neq 0$ and $y \neq 0$, divide by $2x$ and $2y$ respectively to get

$$y^2 = \lambda, \qquad x^2 = \lambda \quad \Longrightarrow \quad x^2 = y^2.$$

With $x^2 + y^2 = 1$, this gives $x^2 = y^2 = \frac{1}{2}$, so $|x| = |y| = \frac{1}{\sqrt{2}}$. Then $f = \left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{1}{4}$. Thus $f_{\min} = 0$ at $(\pm 1, 0), (0, \pm 1)$ and $f_{\max} = \frac{1}{4}$ at the four points $(\pm\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{2}})$.

(vi) $f(x, y) = \sqrt{xy}, \quad g(x, y) = x + y - c = 0.$

Solution. We consider the natural domain $xy \geq 0$; in typical applications one also assumes $x, y \geq 0$ and $c > 0$. On $x, y > 0$,

$$f_x = \frac{1}{2}\sqrt{\frac{y}{x}}, \qquad f_y = \frac{1}{2}\sqrt{\frac{x}{y}}, \qquad \nabla g = (1, 1).$$

The Lagrange conditions say

$$\frac{1}{2}\sqrt{\frac{y}{x}} = \lambda, \qquad \frac{1}{2}\sqrt{\frac{x}{y}} = \lambda.$$

Equating these gives

$$\sqrt{\frac{y}{x}} = \sqrt{\frac{x}{y}}.$$

Squaring both sides,

$$\frac{y}{x} = \frac{x}{y} \quad \Longrightarrow \quad y^2 = x^2.$$

Since we are considering $x, y > 0$, this yields $x = y$. With $x + y = c$, we obtain $x = y = \frac{c}{2}$. The corresponding value is

$$f_{\max} = \sqrt{\frac{c}{2} \cdot \frac{c}{2}} = \frac{c}{2}.$$

If one restricts to $x, y \geq 0$, the minimum is 0, attained at the endpoints $(0, c)$ and $(c, 0)$.

(vii) $f(x,y) = x^2 + y^2, \quad g(x,y) = x^4 + y^4 - 1 = 0.$

Solution.

$$(2x, 2y) = \lambda(4x^3, 4y^3).$$

If $x = 0$, then $y^4 = 1$ so $y = \pm 1$ and $f = 1$. Similarly if $y = 0$, then $x = \pm 1$ and $f = 1$. If $x \neq 0$ and $y \neq 0$, then

$$2x = 4\lambda x^3 \Rightarrow 1 = 2\lambda x^2, \qquad 2y = 4\lambda y^3 \Rightarrow 1 = 2\lambda y^2,$$

so $x^2 = y^2$. With $x^4 + y^4 = 1$ this gives $2x^4 = 1$, hence $x^2 = \frac{1}{\sqrt{2}}$. Thus $f = x^2 + y^2 = 2 \cdot \frac{1}{\sqrt{2}} = \sqrt{2}$ at the four points $(\pm 2^{-1/4}, \pm 2^{-1/4})$.

Since $|x|, |y| \leq 1$ on the constraint, $x^4 \leq x^2$ and $y^4 \leq y^2$, so

$$1 = x^4 + y^4 \leq x^2 + y^2 = f,$$

giving $f \geq 1$ and showing $f_{\min} = 1$ at $(\pm 1, 0), (0, \pm 1)$. Also by Cauchy–Schwarz, $(x^2 + y^2)^2 \leq 2(x^4 + y^4) = 2$, so $f \leq \sqrt{2}$ and the above points attain the maximum.

(viii) $f(x,y) = x^4 + y^4, \quad g(x,y) = x^2 + y^2 - 1 = 0.$

Solution.

$$(4x^3, 4y^3) = \lambda(2x, 2y).$$

If $x = 0$ then the constraint gives $y = \pm 1$, so

$$f = x^4 + y^4 = 1.$$

Similarly if $y = 0$ then $x = \pm 1$ and $f = 1$. If $x \neq 0$ and $y \neq 0$, then

$$4x^3 = 2\lambda x, \qquad 4y^3 = 2\lambda y.$$

Dividing by $2x$ and $2y$ gives

$$2x^2 = \lambda, \qquad 2y^2 = \lambda,$$

so $x^2 = y^2$. Using $x^2 + y^2 = 1$, we get $x^2 = y^2 = \frac{1}{2}$, and therefore

$$f = x^4 + y^4 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

Hence $f_{\min} = \frac{1}{2}$ at $(\pm\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{2}})$, and $f_{\max} = 1$ at $(\pm 1, 0), (0, \pm 1)$.

(ix) $f(x_1, \ldots, x_n) = x_1 + \cdots + x_n, \quad g(x_1, \ldots, x_n) = x_1^2 + \cdots + x_n^2 - 1 = 0.$

Solution. Using Lagrange multipliers:

$$(1, \ldots, 1) = \lambda(2x_1, \ldots, 2x_n),$$

so for each i,

$$1 = 2\lambda x_i \quad \Longrightarrow \quad x_i = \frac{1}{2\lambda}.$$

Thus all coordinates are equal. Let $x_1 = \cdots = x_n = t$. Then the constraint becomes

$$nt^2 = 1 \quad \Longrightarrow \quad t = \pm\frac{1}{\sqrt{n}}.$$

Therefore

$$f_{\max} = \sqrt{n} \quad \text{at} \quad x_i = \frac{1}{\sqrt{n}}, \qquad f_{\min} = -\sqrt{n} \quad \text{at} \quad x_i = -\frac{1}{\sqrt{n}}.$$

(x) $H(p_1, \ldots, p_n) = -\sum_{i=1}^{n} p_i \log_2 p_i$ with $p_i \geq 0$, subject to $p_1 + \cdots + p_n = 1$.

Solution. For $p_i > 0$, set up the Lagrangian

$$\mathcal{L}(p, \lambda) = -\sum_{i=1}^{n} p_i \log_2 p_i + \lambda\Big(\sum_{i=1}^{n} p_i - 1\Big).$$

For each i,

$$\frac{\partial \mathcal{L}}{\partial p_i} = -(\log_2 p_i + 1/\ln 2) + \lambda = 0 \quad \Longrightarrow \quad \log_2 p_i = \text{constant}.$$

Hence all p_i are equal: $p_i = \frac{1}{n}$. This gives the maximum entropy

$$H_{\max} = -\sum_{i=1}^{n} \frac{1}{n} \log_2\Big(\frac{1}{n}\Big) = \log_2 n.$$

For the minimum, note that each term $-p_i \log_2 p_i$ is nonnegative for $0 \leq p_i \leq 1$, so

$$H(p_1, \ldots, p_n) \geq 0.$$

Equality holds at an extreme point of the simplex, for example

$$(p_1, \ldots, p_n) = (1, 0, \ldots, 0),$$

where

$$H = -(1) \log_2 1 = 0.$$

Thus $H_{\min} = 0$ and $H_{\max} = \log_2 n$.